Fatima Zahra Houqqa



AI & Data Science Engineering Student

Machine Learning | Computer Vision | Generative AI | MLOps

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Portfolio: houqqa-fatimazahra.github.io | **GitHub: github.com/houqqa-fatimazahra**

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SUMMARY

Final-year engineering student in Artificial Intelligence and Data Science at EMSI (Rabat, Morocco). I build end-to-end AI solutions: **LLM** assistants with function calling and **RAG**, **Deep Learning** models for **Computer Vision** (PyTorch), and **MLOps** deployment (MLflow, DVC, Docker, Kubernetes). Focused on rigorously evaluated, explainable models (SHAP, Grad-CAM). Seeking an **end-of-studies internship in AI / Machine Learning** for 2026–2027.

PROFESSIONAL EXPERIENCE

AI Intern: LLM assistant for IT asset management | July – August 2026 (2 months)

Secretariat of State for Handicrafts and Social and Solidarity Economy, Rabat, Morocco

- Developed an LLM-based conversational assistant that answers employees' questions about equipment, warranties and stock.
- Connected the LLM to a FastAPI backend through function calling to ground answers in real data and reduce hallucinations.
- Trained a Random Forest model to prioritize hardware renewal (R^2 0.94, MAE 4.23 on synthetic data) and explained its predictions with SHAP.
- *Tech stack: Python, LLM, FastAPI, SQLite, SQLAlchemy, Scikit-learn, Random Forest, SHAP.*

Web & SEO Intern (remote) | 2024

LUZINE, web development and UI/UX design agency, France

- Optimized website content for search engines (SEO) and prepared visuals for the agency's project portfolio.

PROJECTS – code: github.com/houqqa-fatimazahra/tp-mlops · details: houqqa-fatimazahra.github.io

End-to-end MLOps pipeline for a fraud-detection model | EMSI, MLOps course, 2026

- Covered the full lifecycle, from data to production: **reproducible** pipeline, **tracked** experiments, **containerized** deployment.
- Versioned data and pipeline with DVC; tracked experiments and registered models with MLflow; compared AutoML (FLAML, H2O); orchestrated with Prefect, including an explainability check (SHAP).
- Served the model through a FastAPI service in Docker, deployed on Kubernetes: pod self-healing and a rolling update with zero failed requests (0 of 394). *Tech stack: DVC, MLflow, FLAML, H2O, Prefect, SHAP, FastAPI, Docker, Kubernetes.*

AI agent + RAG for personalized prospecting emails (n8n) | 2026, in progress

- Designed the architecture of an AI agent (LLM + RAG) that drafts personalized emails from a knowledge base, with no invented prices or facts.
- Specified quality guardrails: JSON format validation, length checks, amount verification and human approval before sending (implementation in progress). *Tech stack: n8n, LLM, RAG, Google Sheets, Gmail.*

Brain tumor segmentation on MRI (BraTS 2020) | EMSI

- Implemented a multi-class U-Net on 4 MRI modalities, with a patient-level data split to prevent leakage.
- Working prototype trained on a subset of the dataset and evaluated with Dice, IoU and a confusion matrix: Dice ~0.80 on validation and ~0.53 on test. *Tech stack: Python, PyTorch, U-Net.*

Car damage detection and repair cost estimation | University Deep Learning project, 2026

- Trained an EfficientNet-B0 with transfer learning (91% accuracy, 0.91 F1 on validation) and explained its predictions with Grad-CAM.
- Built a Flask application combining detection, cost estimation and a help chatbot. *Tech stack: Python, PyTorch, OpenCV, Grad-CAM, Flask.*

Date fruit image classification with CNN | ESIAS, Specialized Technician final project, 2025

- Built a MobileNetV2 model in a team of 3; handled preprocessing, training, evaluation and the user interface; project defended in 2025. *Tech stack: Python, Keras, TensorFlow, OpenCV.*

Secure e-voting system | FST Errachidia, Bachelor's final project

- Designed, as a team, a voting system based on RSA encryption and blockchain (transparency, fraud prevention).

TECHNICAL SKILLS

Machine Learning & Deep Learning	**Python**, PyTorch, TensorFlow, Keras, Scikit-learn, Random Forest, CNN, U-Net, SHAP
Generative AI	LLM, function calling, RAG, AI agents, conversational assistants
Computer Vision	OpenCV, EfficientNet, MobileNetV2, Grad-CAM, medical image segmentation
MLOps & Big Data	MLflow, DVC, Docker, Kubernetes, Prefect, AutoML (FLAML, H2O), Apache Spark
Backend & APIs	FastAPI, Flask, REST APIs, SQLAlchemy, SQLite, Laravel, Django
Languages & Data	Python, Java, SQL, PHP, JavaScript, Pandas, NumPy, Power BI
Web & Modeling	HTML5, CSS3, Bootstrap, WordPress, UML, Merise, software architecture, networking

EDUCATION

2025 – present	**Engineering degree (5th year), Artificial Intelligence and Data Science**, EMSI, Rabat
2024 – 2025	Bachelor's degree in Software Engineering (in parallel), FST Errachidia
2023 – 2025	Specialized Technician Diploma in Software Development, ESIAS
2020 – 2023	DEUST in Mathematics, Computer Science and Physics (2-year university diploma), FST Errachidia
2019	Baccalaureate, Lycée Mixte, Missour

CERTIFICATIONS

Oracle AI Database Certified Foundations Associate – Oracle, 2026 · **Introduction to Big Data** – UC San Diego, 2026
Agile Project Management – Google, 2026

LANGUAGES

Arabic (native) · French (advanced)
English (B2 – Upper Intermediate) · German (basic)

INTERESTS

Football, table tennis (regional tournaments),
travel, reading.